

Raman Kumar Jha

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Education

Texas A&M University

Doctor of Philosophy in Computer Engineering,

Relevant Coursework: Multi Robot Systems, Deep Learning, Multimodal Computer Vision

College Station, USA

Aug 2026 – May 2030

New York University

Master of Science in Computer Engineering, GPA 3.7/4

Relevant Coursework: Foundation of Robotics, Robot Perception, Reinforcement Learning & Optimal Control for Robotics, Embodied Learning & Vision, Deep Learning; Plan, Learn & Control Autonomous Space Robots

New York, USA

Aug 2024 – May 2026

Savitribai Phule Pune University

Bachelor of Engineering in Computer Engineering, GPA 8.8/10

Relevant Coursework: Principles of Programming Languages, Data Structures & Algorithms; Discrete Mathematics, Artificial Intelligence, Machine Learning, Data Science & Big Data Analytics, Deep Learning, Image Processing

Pune, India

Jul 2019 – Jun 2023

Technical Skills

Programming: Python, C/C++, Matlab, Octave, $\text{\LaTeX}$

ML/AI Frameworks: PyTorch, TensorFlow, OpenCV, Scikit-Learn, Keras, Stable Baselines3

Simulation & Tools: RViz, Gazebo, Isaac, Habitat, Linux, Git/GitHub, Docker, Anaconda, HuggingFace

Robotics & Perception: ROS, SLAM, PCL, Sensor Fusion, Camera Calibration, Motion Planning

LLM & NLP: LangChain, PEFT (LoRA/QLoRA), RAG, NLTK, Prompt Engineering, Multimodal Learning

Data Analysis: NumPy, Pandas, Matplotlib, SciPy, Seaborn

Cloud, and MLOps: AWS, GCP, Azure, Hugging Face, Wandb

Thesis

[1] **Enhancing Spatial Autonomy: Visibility-Consistent Semantic Mapping in Dynamic Environments**

Raman Kumar Jha

Master's Thesis, 2026 [Thesis](#)

Publications

[1] **VCS-SLAM: Visibility-Consistent and Surface-Coupled Semantic SLAM**

Raman Jha, Shuaihang Yuan, Yi Fang.

Robotics and Automation Letters(RAL) 2026 [Paper](#)

[2] **Adaptive Keyframe Selection for Scalable 3D Scene Reconstruction in Dynamic Environments.**

Raman Jha, Yang Zhou, Giuseppe Loianno.

International Conference on Robotics, Computer Vision and Intelligent Systems (ROBOVIS) 2026, [Paper](#).

[3] **RT-X NEt: Cross Attention Network using RGB and Thermal Images for Low-Light Image Enhancement.**

Raman Jha, Adithya Lenka, Mani Ramanagopal, Aswin C. Sankaranarayanan, Kaushik Mitra.

International Conference on Image Processing (ICIP) 2025, [Paper](#).

[4] **A Contrastive Meta-Learning Approach with Isotropic Sparse Decomposition for Scalable Audio-Visual Learning.**

Dhruv Dixit, Paritosh Pandey, Raman Jha, Pranshul Bhatnagar, Shashank Mouli Satapathy.

International Conference on Computing, Communication and Learning (CoCoLe) 2024, [Paper](#).

[5] **Detection of Vehicle Lights and Classification of Its State.**

Raman Kumar Jha

International Conference on Advanced Computing Technologies and Applications (ICACTA) 2023, [Paper](#).

[6] **Prediction of Bankruptcy of a Company Using Machine Learning Techniques.**

V. Kothuru, Raman Kumar Jha, S. Ranjit, B. V. Kumar Reddy, S. Roy, S. Sudheer.

International Conference on Electronics and Sustainable Communication Systems (ICESC) 2022, [Paper](#).

Experience

Embodied AI and Robotics Lab, New York University

Research Assistant(MS Thesis)

New York, USA
Aug 2025 – May 2026

- Mentored by [Shuaihang Yuan](#), and advised by [Prof. Yi Fang](#) for MS thesis.
- Addressed robot object goal navigation and semantic mapping challenges in visually complex environments.
- Developed a physically consistent semantic mapping module to mitigate semantic ghosts and occlusion leakage.
- Implemented a visibility-aware model and surface-coupled regularization to align semantic boundaries.
- Enforced temporal consistency and visibility gating to reduce the Occlusion Leakage Rate (OLR) and enhance spatial reasoning for robust autonomous navigation.

Agile Robotics and Perception Lab, New York University

Research Assistant

New York, USA
Oct 2024 – July 2025

- Mentored by [Yang Zhou](#) and supervised by [Prof. Giuseppe Loianno](#).
- Worked on adaptive keyframe selection for scene reconstruction in dynamic environments.
- Developed a Transformer-based architecture to regress 3D point maps directly from 2D images.
- Optimized keyframe selection by implementing an error-based metric and momentum-based keyframe control.
- Reduced the required number of keyframes by 5% while improving the system's reconstruction on a drone.

Computational Imaging Lab, IIT Madras

Research Associate

Chennai, India
Aug 2023 – June 2024

- Advised by [Prof. Kaushik Mitra](#) and [Prof. Aswin C. Sankaranarayanan](#).
- Developed a cross-attention network for low-light image enhancement and restoration in autonomous systems.
- Used thermal images as a guide for RGB images to improve the performance of the enhancement algorithm.
- Curated and released the V-TIEE dataset, consisting of 50 paired real-world RGB-Thermal sequences.
- Achieved 2 dB performance increase over traditional RGB-only networks for low-light image enhancement.

SeiSei.ai

Founding Machine Learning Intern

Jaipur, India
April 2023 – July 2023

- Designed a Generative AI pipeline for automated video transformation, visual face dubbing, and voice conversion.
- Created a system to detect and extract the landmarks from the face of a particular individual in a video.
- Implemented a facial landmark detection system to drive a GAN-based FreeVC model for voice cloning and a DInet model for visual dubbing.
- Improved inference performance by 17% and successfully integrated the pipeline into the company's production.

Dalhousie University

Mitacs Globalink Research Intern | Research Report

Halifax, Canada
July 2022 – Oct 2022

- Supervised by [Prof. Andrew Rutenberg](#).
- Worked on classifying health variables into low and high dimensions for predicting aging and death.
- Used Variational Autoencoders and Recurrent Neural Networks for model development and testing.
- Trained and tested on the [English Longitudinal Study of Ageing \(ELSA\) dataset](#).
- Developed latent space models improving performance by 14% and designed a modified separate filter to calculate the loss function for each health variable.

Centre of Visual Information Technology Lab (CVIT), IIIT Hyderabad

Applied Research Fellow

Hyderabad, India
Jan 2022 – June 2022

- Supervised by [Prof. Ravi Kiran Sarvadevabhatla](#) and [Prof. C V Jawahar](#).
- Worked on detecting vehicle lights and classifying their states by training multiple object detection models, including YOLOv7, YOLOv5, YOLOv4, Faster R-CNN, and MobileNet.
- Used [Indian Driving Dataset \(IDD\)](#) for training and testing of the detection and classification system.
- YOLO V7 achieved best results with 88.4% mAP and was fine-tuned to improve detection of small lights.
- Applied color thresholding principle for the red and white light recognition in diverse environments.

Projects

Lunar Rover Navigation & Control | [GitHub](#)

Oct 2025 – Dec 2025

- Simulated a six-wheeled lunar rover for autonomous navigation in a GPS-denied, low-gravity environment.
- Targeted robust waypoint traversal while maintaining chassis stability and mitigating risks on uneven terrain.
- Implemented an EKF-based sensor fusion for state estimation and designed a stability-aware proportional control.
- Maintained a 0.6–0.9 stability margin and quantified a 55m localization drift caused by wheel slip in lunar regolith.

Self-Drive: Virtual Integrated Project | [GitHub](#)

Jan 2025 – May 2025

- Planned an autonomous driving system for real-world navigation in complex maze, and in real-world.
- Designed and integrated perception, planning, and control modules for robust decision-making.
- Utilized DINOv2-based lane detection and object recognition, and optimized A* for efficient path planning.
- Achieved 95% navigation accuracy in real-world maze environments with safe and reliable performance.

Diffusion-Based Model Predictive Control | [GitHub](#)

Feb 2025 – May 2025

- Designed a control strategy to improve robotic trajectory optimization in uncertain and dynamic environments.
- Enhanced MPC by integrating probabilistic modeling to handle uncertainty in unseen conditions.
- Combined diffusion models with MPC and implemented probabilistic trajectory generation using diffusion.
- Achieved a 20% improvement in trajectory optimization performance over traditional MPC approaches.

CLIP Model Benchmarking on Flickr8K | [GitHub](#)

Apr 2025

- Designed CLIP vision-language models for zero-shot image retrieval and classification on Flickr8K dataset.
- Objective was to assess how encoders impact retrieval quality and response to both concrete and abstract prompts.
- Constructed and pretrained three model pipelines, tested on custom prompts to evaluate performance.
- Larger models excelled at retrieval tasks, however smaller ones trained faster but ,performed worse on queries.

LoRA Fine-Tuning for Text Classification | [GitHub](#)

Mar 2025

- Adapted a pre-trained transformer (RoBERTa) for a text classification task in a resource-constrained environment.
- Aimed to maximize classification accuracy while minimizing additional parameter usage using PEFT techniques.
- Fine-tuned RoBERTa with LoRA adapters using low learning rate (1e-4), weight decay, and optimized batch size.
- Achieved 94.06% validation accuracy (final loss: 0.16) and 85.9% Kaggle test accuracy on final competition.

2D Quadrotor Control Using Reinforcement Learning | [GitHub](#)

Oct 2024 – Dec 2024

- Designed a control solution for a 2D quadrotor to reach target positions while avoiding obstacles.
- Implemented a reinforcement learning framework for stable navigation under dynamic constraints.
- Trained a PPO-based policy and engineered rewards balancing target accuracy, stability, and obstacle avoidance.
- Achieved robust flight control with smooth trajectories, minimal oscillations, and reliable obstacle avoidance.

American Sign Language Recognizer | [GitHub](#)

Jun 2022 – Jul 2022

- Built an American Sign Language recognizer using real-time hand gesture detection.
- Applied the YOLOv5 object detection framework to real-time camera images for accurate sign recognition.
- Recognized five distinct American Sign Language signs, including Hello, I Love You, Yes, No, and Thank You.
- Achieved 97.4% recognition accuracy across all American Sign Language sign classes during real-time camera-based testing and evaluation.

Driver Drowsiness Detector | [GitHub](#)

May 2021 – Jun 2021

- Developed a driver drowsiness detector based on real-time eye-state analysis.
- Trained a CNN-based model on real-time camera image sequences to detect signs of drowsiness.
- Classified the driver's state as awake or drowsy using extracted eye condition features and blink patterns.
- Achieved 85.2% detection accuracy for driver drowsiness classification across real-time camera-based testing and evaluation.

Teaching Experience

Teaching Assistant

Spring 2026

Computer Vision for Sciences and Engineering

New York University

- Assisted students with paper readings, research critiques, and developing effective research presentation skills.
- Mentored students through structured project guidance, including proposals and feasibility studies.
- Evaluated weighted presentations, peer-review assignments, and initial and final project reports.

Teaching Assistant

Fall 2025

Advanced Topics in Computer Vision

New York University

- Assisted students with their presentations, paper reviews, project queries, and coursework.
- Conducted office hours for the students to help them with their projects in Computer Vision.
- Graded student presentations, paper reviews, and course projects submitted through Brightspace.

Teaching Assistant

Spring 2025

Physics Informed Machine Learning

New York University

- Mentored students with their homework, quizzes, and projects, and addressed doubts on Piazza.
- Held office hours and tutorial sessions for the students in Physics-Informed Machine Learning.
- Evaluated student homework, quizzes, assignments, and midterms for the course on Brightspace.

Teaching Assistant

Spring 2025

Chemical Engineering Computation

New York University

- Assisted students with their homework, assignments, and in-class doubts on Python programming.
- Held office hours and guided students through their mini projects and final course projects.

- Graded homework, quizzes, assignments, and midterms for Chemical Engineering Computation on Brightspace.

Teaching Assistant

Spring 2024

Modern Computer Vision

IIT Madras

- Conducted tutorial classes on the basics of programming and deep learning for students.
- Evaluated student assignments and projects for the computer vision course through Moodle.
- Mentored students on their course projects, competitions, and hackathon participation.

Teaching Assistant

Spring 2023

Machine Learning

Savitribai Phule Pune University

- Graded student assignments, projects, and exams for the machine learning course.
- Held office hours and assisted students with their homework, projects, and hackathons.
- Assisted students and answered their questions on Sowl about reports and final projects.

Achievements and Extracurriculars

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| 1 Judge , Hackathon LLUS X PULSE hosted by Korean International Students Organization | 04/2026 |
| 2 Winner , Spark Hackathon hosted by NVIDIA, and sponsored by ACER | 04/2026 |
| 3 Outstanding Reviewer , British Machine Vision Conference (BMVC) | 11/2024 |
| 4 Recipient , merit-based scholarship of \$7000 per year from New York University | 04/2024 |
| 5 Finalist , Smart India Hackathon | 08/2022 |
| 6 Recipient , Mitacs Globalink Research Fellowship | 03/2022 |
| 7 Winner , Codeheat contest, FOSSASIA | 09/2021 |
| 8 Recipient , Script Fellowship, Script organization | 08/2021 |

Reviewer Experience

Reviewer: BMVC 2026, NeurIPS 2025, BMVC 2025, ICLR 2024, BMVC 2024, ICVGIP 2023

Leadership and Extra-curriculars

- 1 **Treasurer**, [Hindu Student Union NYU](#) (July 2025 – May 2026). Directed financial operations, including budget forecasting and allocation, while securing resources for diverse events to promote community engagement.
- 2 **Treasurer**, [IEEE NYU Student Branch](#) (Jun 2025 – May 2026). Managed financial activities, budget planning, and coordinated funding for technical events and workshops for the club.
- 2 **Steward**, [GSOC NYU](#) (May 2025 – May 2026). Represented as a steward for NYU Tandon School of Engineering, and helped students with queries, grievances, and issues related to their on-campus jobs.
- 3 **Mitacs Globalink Ambassador**, [Mitacs Globalink Research Internship](#) (Mar 2023 – Jun 2023). Assisted and guided upcoming research interns regarding the Mitacs Globalink Research Fellowship.
- 4 **Machine Learning Lead**, [Google Developer Student Club](#) (Aug 2021 – Jul 2022). Conducted events, seminars, tutorials, and guided juniors for research in Machine Learning.